

Ian Todd
Sydney Medical School
University of Sydney
Sydney, NSW, Australia
itod2305@uni.sydney.edu.au

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Professor A.U. Igamberdiev
Editor, BioSystems

Dear Professor Igamberdiev,

Please find enclosed the revised manuscript BIOSYS-D-25-00981, “Coherence time in biological oscillator assemblies bounds the rate of state registration,” for consideration as a revised submission to *BioSystems*.

I thank both reviewers for their constructive feedback. All points raised have been addressed, and the revision includes substantial additional developments.

Response to Reviewer #1

Reviewer: “The speed of thinking is restricted by the Landauer and Margolus-Levitin bounds. . . This should be discussed in the revised version.”

Response: Section 2.4 (“Quantum and thermodynamic limits of computation”) now explicitly addresses the Landauer bound, the Margolus-Levitin quantum speed limit, and the energy-time uncertainty relation. For neural-scale energies, QSL and Landauer bounds are far below observed cognitive timescales (by approximately eleven orders of magnitude) and are therefore not rate-limiting in this regime. The power limit (τ_{power}) is now quantified with a detailed back-of-envelope calculation (§2.1) showing it is non-binding even at biophysical (not merely Landauer-floor) energy scales.

Response to Reviewer #2

Reviewer: “Speed of thought should be changed to speed of neural processes associated with thinking.”

Response: “Speed of thought” has been replaced with “neural processes associated with cognition” throughout, including the abstract.

Reviewer: “Quantum limits of information transmission based on energy-time uncertainty relation should be discussed.”

Response: Energy-time uncertainty is discussed in §2.4, following the framework developed

by Igamberdiev (2021, 2025, 2026).

Reviewer: Cite four specific BioSystems papers (DOIs provided).

Response: All four are now cited: Igamberdiev (2021, 104471), Igamberdiev (2025, 105451), Igamberdiev (2026, 105707), and Khrennikov et al. (2025, 105573).

Reviewer: “Abstract should not contain references. Also, if formulas cannot be avoided in abstract, it would be better that you simplify them wherever possible.”

Response: The abstract contains no citation numbers. Mathematical notation has been simplified; only simple inline symbols (M , r , ε) are retained for precision.

Additional improvements

Beyond the reviewer requests, the revision includes several substantial developments:

- **Phase delta regime** (§2.3): New section identifying the pre-commit interval—in which structured inter-module phase relationships bias downstream outcomes without discrete state registration—as a computationally productive mode operating below the Landauer threshold. Includes formal Definition 2 and the speed-flexibility trade-off (Proposition 2).
- **Substrate identification** (§2.1): Argument from the framework’s own definitions that the extracellular electric field is the physical carrier of the phase delta regime: it is the only substrate in neural tissue that is simultaneously continuous, spatially extended, and high-dimensional. The causal significance of ephaptic coupling follows from two premises: fields modulate spike timing (Anastassiou et al. 2011), and timing is causally significant (the paper’s central result). The magnitude of the effect scales with effective dimensionality D_{eff} .
- **Power limit quantified** (§2.1): Back-of-envelope calculation showing $\tau_{\text{power}} \sim 10^{-15}$ – 10^{-8} s, non-binding even at biophysical energy scales. Added to the hierarchy listing in §2.4.
- **Improved validation methodology:** Kaplan-Meier survival analysis for censored first-passage times, theory-consistent fitting ($\hat{\alpha} = 0.99$, $R^2 = 0.97$), and regime-boundary diagnostics across three network topologies.
- **New sections:** Frequency-dimensionality relationship (§4.5), subjective temporal scaling (§4.6) with pharmacological predictions across four drug classes from a two-parameter structure.
- **Alpha oscillations and perception sets** (§3.3): Integrated recent work by Wutz (2024) showing alpha reflects internally generated “perception sets.” We show this is complementary: a perception set can be interpreted as a high-dimensional internal template, and the temporal effects documented—dilation under richer processing, compression under impoverished conditions—correspond to the exponential dependence of

τ_{coh} on coordination depth.

- **Synchronisation theory connections** (§4.5, §4.8): Added Buzsáki & Draguhn (2004) for the frequency-hierarchy view (§4.5). Added connections to the EI-Kuramoto model (Kuroki & Mizuseki, 2025) and criticality as a computational operating point (Hengen & Shew, 2025) in the limitations section (§4.8).
- **Expanded limitations:** Parameter identifiability, regime boundaries ($R^2 = 0.28$ for non-modular networks), module-size confound analysis, dwell-time sensitivity.

The manuscript has grown from 41 to 55 pages to accommodate these developments.

Data and code availability. All simulation code and parameter estimation workflows are available at <https://github.com/todd866/coherence-time>.

Sincerely,

Ian Todd
Sydney Medical School
University of Sydney